TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Analysis of Group I (Pb, Ag, Hg)

Separating and confirming the three chloride-group cations — lead by its hot-water solubility, silver by its ammonia complex, mercury by the blackening test.

CHE-1106 · CHEMISTRY-I SESSIONAL · EXPERIMENT 5 · LEVEL 1 TERM 1

1. Aim and the separation logic

Aim: to analyse the Group-I precipitate (the white chlorides from Experiment 4) and identify which of Pb^{2+}, Ag^{+} and Hg_2^{2+} are present. The three chlorides look alike — all white — but each carries one distinguishing behaviour, and the analysis is simply those three behaviours arranged in sequence:

CHLORIDE	DISTINGUISHING BEHAVIOUR
$PbCl_2$	Appreciably soluble in hot water (the others are not)
AgCl	Insoluble in hot water, but dissolves in NH_4OH as the complex $[Ag(NH_3)_2]^+$, and reprecipitates on acidifying
Hg_2Cl_2	Insoluble in both, but turns black with NH_4OH (a mixture of Hg and $HgO{\cdot}HgNH_2Cl$ forms) — the blackening is unique in the group

2. Procedure

1. **Obtain and wash the group precipitate:** original solution + dilute HCl dropwise until no further precipitation; filter; wash with cold water containing a drop of dilute HCl.
2. **Extract lead:** pour hot water repeatedly through the precipitate on the filter paper. Collect the hot filtrate — it contains $PbCl_2$ if lead is present. Divide it:
   (a) Cool one portion: white needle-like crystals of $PbCl_2$ separate.
   (b) To another portion add KI: bright yellow precipitate (PbI_2), dissolving on heating, returning as golden spangles on cooling.
   (c) Or add K_2CrO_4: yellow precipitate ($PbCrO_4$).
3. **Extract silver:** pour dilute NH_4OH through the residue on the paper. If AgCl is present it dissolves. Collect the filtrate and acidify with dilute HNO_3: a white precipitate (AgCl again) confirms silver. (A portion may also be tested: the ammoniacal solution gives no precipitate with HNO_3 only if silver is absent.)

4. **Test the residue for mercury:** the residue left on the paper after the ammonia treatment is examined: if it has turned **black** during the NH_4OH treatment, Hg_2^{2+} is present (the ammonia both fails to dissolve Hg_2Cl_2 and blackens it). Confirm further in a tube: the black residue dissolves in aqua regia; the solution, diluted, gives an orange/yellow precipitate with NaOH or a scarlet precipitate with KI (in exact quantity) — the standard mercury confirmations as your laboratory directs.

3. Flow summary

Group-I precipitate → (hot water) → filtrate = **Pb^{2+}** (KI/chromate confirms) | residue → (NH_4OH) → filtrate, acidified = **Ag^{+}** (white ppt returns) | final residue blackening with NH_4OH = **Hg_2^{2+}**. One precipitate, three doors, each opening for one cation only.

4. Notes and interferences

- $PbCl_2$ is slightly soluble even in cold water, so lead commonly appears again in Group II (as black PbS) — finding lead in both groups is normal, not an error.
- The ammonia extract must be acidified promptly and completely; an under-acidified filtrate keeps silver dissolved and reports a false negative.
- Mercury compounds are poisonous: residues go to the laboratory's heavy-metal waste, never the sink, and hands are washed after the experiment — a working rule for every mercury test in this course.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. The Group-I precipitate is white. On treatment with hot water part dissolves; the hot solution on cooling gives white needles, and with KI a yellow precipitate. Name the cation and write both evidences.
2. How is AgCl separated from Hg_2Cl_2, and how is the dissolved silver proved to be silver?
3. What does blackening of the residue with NH_4OH indicate, and what is the black material (in words)?
4. Why may lead be found again in Group II, and in what form?
5. A student acidifies the ammoniacal filtrate only slightly and sees no precipitate, reporting silver absent. Criticise the procedure.
6. State the safety rule for mercury residues and its reason.

Answer key

1. Pb^{2+}. Evidences: $PbCl_2$'s hot-water solubility with recrystallisation as white needles on cooling; PbI_2's bright yellow precipitate with its golden-spangles behaviour on heating and cooling.
2. NH_4OH dissolves AgCl as $[Ag(NH_3)_2]Cl$ while Hg_2Cl_2 stays undissolved (and blackens). Silver is proved by acidifying the ammoniacal filtrate with HNO_3: white AgCl reprecipitates as the complex is destroyed.
3. Hg_2^{2+} (mercurous). The black colour is a mixture produced from Hg_2Cl_2 by ammonia — finely divided mercury metal together with a white mercury–amido compound, appearing black overall.
4. Because $PbCl_2$ is appreciably soluble in cold water, some lead escapes Group I into the filtrate and precipitates in Group II as black PbS with H_2S.
5. The complex $[Ag(NH_3)_2]^+$ breaks down only in properly acidic solution; "slightly acidified" leaves ammonia unconsumed and the silver dissolved — the negative is unproved. Acidify until the solution is distinctly acid to litmus, when AgCl must fall if silver is present.
6. Mercury residues go to the designated heavy-metal waste, never to the sink or bin, and hands are washed — mercury compounds are cumulative poisons and environmental contaminants.

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